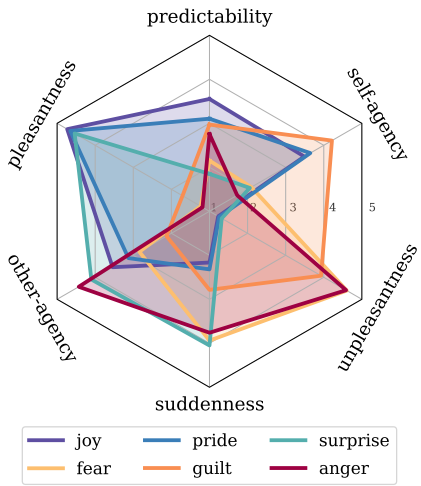

Figure 5: Appraisal emotion associations extracted from the dataset. For example, when participants report *anger*, they typically perceive a high degree of *other-agency* and a low level of *pleasantness* in the situation.

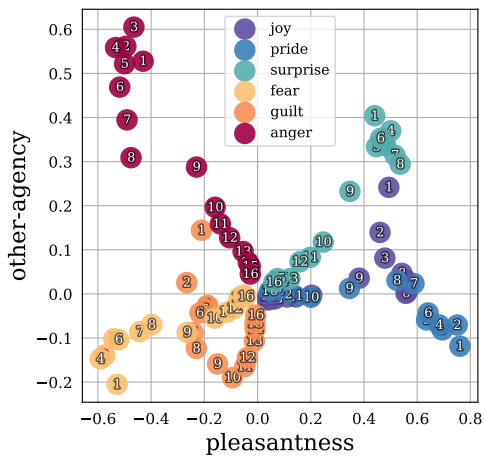

Figure 6: Cosine similarity of emotion vectors with *pleasantness* and *other-agency* appraisal vectors extracted from the hidden state of different Llama 3.2 1B layers. Layer number is written inside each marker.

8 Emotion-Appraisal Mappings

In this section, we analyze the representations of emotions and appraisals to reveal a structure within the latent LLM representations. Remember the weight matrix $\mathbf{W} \in \mathbb{R}^{d \times C}$ introduced in Section 5. Let $\mathbf{w}_e \in \mathbb{R}^d$ represent the column e of $\mathbf{W}$ corresponding to the emotion index $e \in \{1, \cdots, C\}$. We define the cosine similarity of appraisal a with emotion e as $sim(a,e) = \frac{\mathbf{v}_a^\top \mathbf{w}_e}{\|\mathbf{v}_a\|_2 \|\mathbf{w}_e\|_2}$.

Figure 6 shows the similarity score of emotion vectors with two appraisal vectors, i.e. the *pleasantness* and *other-agency*, throughout Llama 3.2 1B layers. Notably, we observe psychologically plausible appraisal-emotion mappings across all layers. However, the projection strength peaks in the early layers and fades to near zero in the final layer, suggesting orthogonality in the final layers.

We hypothesize that in the earlier layers, there exists a meaningful structure capturing appraisal and emotion concepts, which aligns with our expectations from the appraisal theory. However, these concepts gradually decouple as the processing progresses through the network, and by the final layers, they become orthogonal and fully decoupled, reflecting the specialization of the network toward higher-level tasks. We finish this section by drawing the connection to our findings in previous sections. Notice that the decoupling starts around the critical layers, e.g. layers 9 and 10 in Llama 3.2 1B, which we identified in previous sections. Therefore, our observations suggest that *the appraisals build a foundation to understand emotion representations in LLM hidden states, but the structure vanishes as we progress through the network.*

9 Intervention on Appraisal Concepts

After finding the appraisal vectors, we investigate the possibility of indirectly modifying the emotion of an input example by modulating its appraisals within the model representations. For this purpose, we need to isolate the role of each appraisal a, by considering its associated latent vector $\mathbf{v}_a$ and distinguishing it from other appraisal vectors. More precisely, we define $\mathbf{V}_{-a} := [\mathbf{v}_1, \ldots, \mathbf{v}_{a-1}, \mathbf{v}_{a+1}, \ldots, \mathbf{v}_n]$ by concatenating all appraisal vectors except $\mathbf{v}_a$. Next, we introduce the *unique effect vector* of appraisal a as $\mathbf{z}_a := (\mathbf{I} - \mathbf{P}_{-a})\mathbf{v}_a$, where $\mathbf{P}_{-a} = \mathbf{V}_{-a}(\mathbf{V}_{-a}^\top \mathbf{V}_{-a})^{-1}\mathbf{V}_{-a}^\top$ is the projection matrix onto the column space of $\mathbf{V}_{-a}$ and $\mathbf{I} \in \mathbb{R}^{d \times d}$ is the identity matrix. We perform *appraisal modulation* by injecting $\mathbf{z}_a$ into the model's latent representation. More formally, the intervention is expressed as:

$$\mathbf{x} \leftarrow \mathbf{x} + \beta \frac{\mathbf{z}_a}{\|\mathbf{z}_a\|_2},$$

where $\mathbf{x}$ on the RHS is the original latent representation, e.g., a hidden state vector from a specific layer, and β is a scaling factor controlling the strength of the concept modulation. Notice that a positive β corresponds to an appraisal promotion while a negative β has the opposite effect of appraisal demotion. To measure the success of interventions, we evaluate the new emotion label

derived from this modification and repeat this procedure across all examples. Notice that, this modification is applicable to any layer of the model.

Figure 7 illustrates concept modulation results with different magnitudes of β on layer 9 of Llama 3.2 1B. Notably, we observe a remarkable alignment with theoretical and intuitive expectations. For instance, increasing the *pleasantness* appraisal promotes both *joy* and *pride*, aligning with the fact that both of these emotions have high associations with the *pleasantness* appraisal.

In contrast to these results, the appraisal modulation when applied to earlier layers, does not generate psychologically valid results and is totally ineffective when applied to later layers. This observation matches the intuitions on the mechanism we provided earlier; Intervening on the early layers is not valid since modifications to latent representations are overwritten by emotion-specialized mid-layers. On the other hand, intervention on final layers is not effective because of the orthogonality of concepts as demonstrated in Section 8. Appendix F presents the full results, including intervention across all layers of Llama 3.2 1B.

Appraisal theory is also predictive of the situations in which two appraisals are promoted simultaneously. To test this capability in LLMs, we perform an intervention on the superposition of two appraisal dimensions: *other-agency* and *pleasantness*, with mathematical details provided in Appendix F. The results, depicted in Figure 7, show a successful promotion of emotion *pride* with no further occurrences of *joy*. These findings provide strong evidence that layer 9 in Llama 3.2 1B directly contributes to cognitive processes related to emotions (see Appendix F for experiments with other appraisal concepts). Additionally, we provide complementary experiments such as direct emotion promotion using emotion vectors in Appendix D.

10 Discussion

We employed mechanistic interpretability techniques to investigate the inner workings of emotion inference in LLMs. Our results reveal that mid-layer MHSA units within these models are responsible for processing emotional content. By applying linear algebraic manipulations to modulate the antecedents of emotions, i.e. the appraisal concepts, we steered the model outputs in controlled and predictable ways. This is particularly important for ensuring the *reliability and steerability* of LLMs in high-stakes affective domains such as legal decision-making and clinical therapy.

A key distinction of our work is that we grounded our experiments on psychological theory and applied MI analysis on *in-the-wild examples*, rather than relying on *synthetically generated simplistic structures*, as seen in prior studies (Merullo et al., 2024). For example, Wang et al. (2023a) study the Indirect Object Identification task, by considering a fixed input structure, such as *"person1 and person2 had fun at school. person2 gave a ring to"* where the model is expected to predict *"person1"*. Hanna et al. (2023) study the "Greater-than" task using a dataset of examples like *"The war lasted from 1517 to 15"*, where the model is expected to predict any two-digit number larger than 17. While such notable efforts have demonstrated the feasibility of identifying end-to-end specialized circuits for these tasks, effectively interpreting them still requires structured inputs, making it challenging to generalize findings to more naturalistic settings.

Our work also highlights new opportunities for future research. Despite significant advancements in understanding human emotions, debates persist regarding the definition of emotion, the role of cognition in emotion, and the mechanisms underlying emotion inference (Ortony et al., 2022; Ellsworth and Scherer, 2003; Moors, 2013; Barrett, 2017). In parallel, cognitive neuroscience has explored the neural basis of emotion in support of differing theoretical perspectives (Kragel et al., 2024). The study of LLMs, combined with insights from emotion theory and neuroscience, opens a unique intersection for advancing our understanding of emotions (Sievers and Thornton, 2024).

Furthermore, our steering approach opens promising possibilities for conditioning LLMs to exhibit specific personality traits or moods, which could benefit applications requiring tailored affective responses (Jiang et al., 2023, 2024a; Petrov et al., 2024; Suh et al., 2024; Li et al., 2024; Suh et al., 2024). However, to ensure these interventions do not introduce unintended disruptions to other critical language-processing functions, it is essential to rigorously evaluate models on standard NLP benchmarks after inducing traits or moods.

Given LLMs' increasing societal impact —spanning areas such as mental health, legal decision-making, and human-AI interaction—it is imperative to deepen our understanding of their internal mechanisms. Our study breaks new ground in the

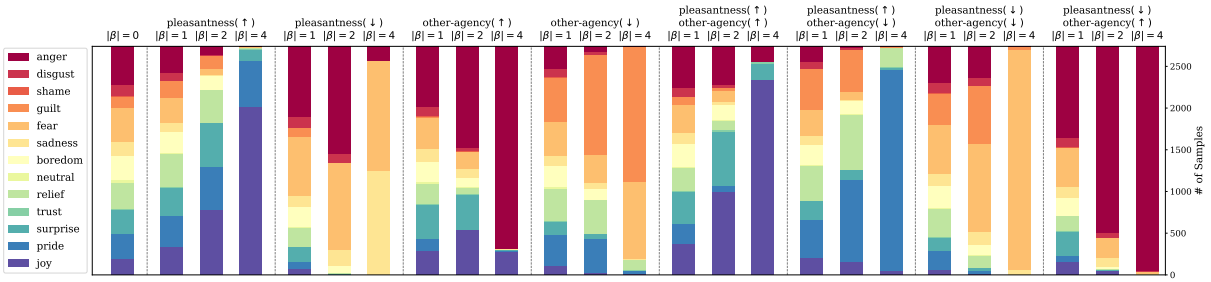

Figure 7: Results of appraisal concept modulation by intervening at layer 9 hidden states of Llama 3.2 1B for increasing scaling factors (β). $|\beta| = 0$ represents the original distribution of emotions in the dataset. For example, promoting (↑) or demoting (↓) *other-agency* significantly increases the share of *anger* and *guilt*, respectively. Similarly, promoting or demoting *pleasantness* increases the share of *joy*/*pride* and *sadness*/*guilt*/*anger* outputs, respectively. Additionally, promoting *pleasant other-agency* significantly increases the share of *joy* outputs, while the promotion of *unpleasant other-agency* significantly increases the share of *anger*.

interpretability of emotion inference in LLMs, offering a novel way to causally intervene in emotional text generation. These findings hold promise for improving safety and alignment in sensitive affective domains. Moving beyond black-box approaches to rigorously test and refine LLM emotion processing will not only advance the field of LLM interpretability but also unlock new pathways for more responsible AI systems.

11 Limitations

In this study, we build upon the linear representation hypothesis (Mikolov et al., 2013a,b; Levy and Goldberg, 2014; Elhage et al., 2022)—the idea that high-level concepts are encoded linearly within model representations (Park et al., 2024). This hypothesis is particularly appealing because, if true, it could enable simple and effective methods for interpreting and controlling LLMs—an approach we leveraged to localize and manipulate latent emotion representations. However, despite notable efforts to formalize the notions of linearity (Park et al., 2024) and orthogonality (Jiang et al., 2024b) in model representations, recent research suggests that not all features are encoded linearly (Engels et al., 2025). Further investigation is needed to improve clarity and robustness in this area.

Furthermore, we demonstrated the ability to manipulate affective outputs by modifying appraisal concepts. Nevertheless, the precise nature of this relationship remains unclear—it is possible that appraisals are merely correlated with emotions rather than exerting a direct causal influence or that the relationship follows an inverse causal pattern. Establishing causality requires further investigation in future studies to disentangle directional dependencies. A deeper understanding of the interplay between LLM emotional inference, emotion theory, and neuroscience will be crucial for both theoretical insights and practical applications. Addressing these challenges will refine our understanding of LLMs and enhance their reliability in affective computing.

12 Ethical Impact Statement

This study re-analyzes previously collected, de-identified data that had already undergone ethical review. The dataset is used for investigating the inner mechanisms by which auto-regressive LLMs process emotion. However, caution must be exercised when generalizing these results to models not examined in this work, to superficially similar tasks, or to different languages.

Our analysis highlights potential concerns for those deploying LLMs in high-stakes affective domains or for generating emotionally charged content. Given the risks associated with emotional manipulation by LLMs, it is crucial to develop a deeper understanding of how these models process emotions. To this end, we advocate for further research in this domain to ensure that LLMs align with ethical standards and human-centered AI principles.

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